Raunak Singh

Atlanta, GA 30332 | 5512289483 | raunak.singh.inventor@gmail.com | US Permanent Resident

Objective

Computer Engineering student at Georgia Tech with USACO Gold distinction and experience building high-performance computing and C++ systems. Seeking to apply experience with software development and embedded systems to an internship in Summer 2027.

Education

Georgia Institute of Technology | Atlanta, GA *August 2026 – Present*
Bachelor of Science in Computer Engineering Expected Graduation, June 2030

Skills

Programming: C/C++, Java, Python, SQL, JavaScript/React.
Platforms: Linux (Ubuntu, Petalinux, Debian), Unix
Hardware: ESP32, ARM mbed microcontroller, Arduino, Raspberry Pi, oscilloscope
Software: CLion, Intellij, VSCode, ICECube2, GitHub
Languages: English (fluent), Punjabi (fluent), Hindi (fluent)

Experience

Aptimized LLC | Wayne, New Jersey *September 2025 – June 2026*
IT Intern
Aptimized LLC provides expert IT consulting, digital transformation, and ERP/HCM solutions to modernize enterprise technology and scale operations.

- Built REST APIs in ASP.NET Core and SQL Server for a client-facing staffing and incident-management platform used by enterprise customers
- Created client-facing PowerPoint process-flow diagrams for incident management, payroll, and change-management proposals, translating technical workflows into clear stakeholder presentations

Princeton University Puchalla Biophysics Lab | Princeton, New Jersey *Summer 2025*
Researcher

- Built 3D electromagnetic-field simulation pipeline, replacing an inaccurate 2D workflow
- Scaled the pipeline to 5,000 cores on Princeton's Stellar Red Hat Linux HPC cluster, automating 850+ Elmer FEM numerical solver jobs with Bash and SLURM; improved agreement with experimental measurements by 55%.

Projects

Duke's Garden: Java 26 AI-backed Buddy for Home Gardeners *Fall 2026*
Full-stack Java Spring Boot website using Vaadin and PostgreSQL for houseplant-care management.

- Implemented GenAI-backed plant identification, disease analysis, and contextual Plant Assistant guidance through Spring AI and an OpenAI-compatible API.
- Built an authenticated REST workflow connecting Raspberry Pi devices for sensor-triggered care-event logging

BikeWatch: Protecting Pedestrians with Xilinx Kria KV260
Deployed a deep-learning model on a DPU to alert pedestrians when a bicycle approaches

- Built a C++ inference pipeline on a Xilinx Kria KV260 running PetaLinux/Vitis AI to detect pedestrians from live camera input.

Leadership or Activities

Bergen County Academies Competitive Programming Team | Captain *September 2023 – June 2026*

- Grew the overall team 5× and led contest preparation through hosting 24 original contests and 30+ practice sessions

Awards

- **U.S.A Computing Olympiad Gold:** Rank #1 in 2025 February Silver & Rank #1 in 2024 January Bronze contests
- **Won 3rd in UPenn PClassic**
- **Won 6th in Lockheed Martin CodeQuest**
- **UC Berkeley CALICO 2x Bronze**